cs_213[1]

howdy

question 1

1. Convert the following hexadecimal numbers to binary.

   0x1010

0b		?

   0x9ea5d3b

0b		?

2. Evaluate the following expressions, giving your answer as a hexadecimal number.

   0x03 + 0x8a

0x	hex number	?

   0xcb + 0x4e

0x	hex number	?

   0x9f52 - 0x20a

0x	hex number	?

question 1

- every hexit corresponds to (at most) 4 binary digits,
  - map each hexit to their corresponding binary representations

0x4 -> 0b0100

1. Convert the following hexadecimal numbers to binary.

0x1010

0b

0x9ea5d3b

0b

question 1

- add/substract column by column
  - helpful to think of these in decimal
- carry = 0xf

2. Evaluate the following expressions, giving your answer as a hexadecimal number.

   0x03 + 0x8a

   0x | hex number

   0xcb + 0x4e

   0x | hex number

   0x9f52 - 0x20a

   0x | hex number

question 2

Assume that `a` is stored starting at address `0x1000`. What are the values of the bytes from address `0x1000-1003` for big- and little-endian representations? Write `N/A` if the values are not known. Treat each subquestion independently.

1.
```
int a = 0x80;
```

Big endian:

0x1000		?	0x1001		?	0x1002		?
0x1003		?						

Little endian:

0x1000		?	0x1001		?	0x1002		?
0x1003		?						

question 2

- little endian: least significant bit/byte first
- big endian: most significant bit/byte first

Assume that `a` is stored starting at address `0x1000`. What are the values of the bytes from address `0x1000-1003` for big- and little-endian representations? Write `N/A` if the values are not known. Treat each subquestion independently.

1. `int a = 0x80;`

Big endian:

0x1000		0x1001	
0x1002		0x1003	

Little endian:

0x1000		0x1001	
0x1002		0x1003	

question 3

Evaluate each of the following pieces of Java code to determine what they output.

```java
int a = 0x7f << 12;
System.out.printf("%x", a);
```

hex number

```java
int a = ((byte)0x7f) << 12;
System.out.printf("%x", a);
```

hex number

```java
int a = ((byte) 0xae) << 16;
a = a & 0x00ff0000;
System.out.printf("%x", a);
```

hex number

```java
int a = ((byte) 0xae) << 16;
a = a | 0x00ff0000;
System.out.printf("%x", a);
```

hex number

question 3

- shifting by 4 bits = shift by 1 hexit
- beware of sign extension!!

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hex number

```java
int a = ((byte) 0xae) << 16;
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hex number

wooo you finished your first 213 lab!!
thanks for coming!